

Accurate prior of binomial distribution with Bernstein polynomial and its Bayesian inference

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Abstract: Data defined on the discrete set $n = \{0, 1, \dots, m\}$ widely exists in biomedicine, finance, engineering, social sciences and other fields. As the basic model for fitting such data, binomial distribution has the advantages of simple mathematical form and clear statistical significance, but its inherent assumptions lead to limited fitting effect for the over-dispersion phenomenon that is common in the real data. The Bayesians reconstructs the success probability p of binomial distribution into a random variable, and uses beta distribution as a prior to derive beta-binomial distribution. The model not only overcomes the fitting defect of binomial distribution to over-dispersion data, significantly improves the robustness, but also has good parameter interpretation, which has become a popular tool in various fields. However, the subjective choice of its prior distribution may lead to the deviation of prior information representation, which is difficult to adapt to the diversity characteristics of complex real data. This paper proposes an accurate prior framework for constructing binomial distribution success probability based on Bernstein polynomial, which can flexibly adapt to all kinds of prior information characteristics, and establishes the corresponding theoretical system of statistical inference and Bayesian inference. Simulation experiments and real data applications verify the superiority and practicability of the proposed distribution and method.

Keywords: Bayesian inference, Bernstein polynomial, Binomial distribution, MM algorithm, Quantile regression model.

1. Introduction and Background

1.1 Literature review

Data defined on the finite discrete set $\mathbb{N}_{0:m} \triangleq \{0, 1, \dots, m\}$ widely exists in empirical research, and its essence is the counting observation constrained by the upper bound m . This kind of data is universal in biomedical fields (number of gene mutation sites detected), industrial engineering (number of product defects under fixed sampling) and social sciences (frequency statistics of discrete scoring options). The core value is to describe the mechanism of random events under the framework of limited resources. For instance, clinical trials need to evaluate the number of effective cases in a fixed sample size, and social surveys need to quantify the group tendency within the preset options. With the wide application of high-dimensional discrete data (such as gene expression matrix of single cell sequencing), the demand for theoretical deepening and model innovation of finite discrete distribution is increasingly prominent.

As the cornerstone of modeling finite discrete data, $\text{Binomial}(m, p)$ distribution describes the probability distribution of the number of successful events x in m independent Bernoulli trials, whose *probability mass function* (pmf) is

$$\text{Binomial}(x|m, p) = \binom{m}{x} p^x (1-p)^{m-x}, \quad x \in \mathbb{N}_{0:m}.$$

The core assumption of the model is the test independence and constant success probability p , which makes it a standard tool in quality control, medical statistics and other fields. However, the actual data often violate their ideal assumptions: first, the phenomenon of over-dispersion (the observation variance is significantly greater than $mp(1-p)$) is widespread in the actual data. Secondly, the homogeneous p hypothesis is difficult to adapt to heterogeneous groups. Some traditional improvement methods, such as quasi-binomial model, which adjust variance by introducing dispersion parameters ϕ , but fail to solve the essential defects of probability structure. And the generalized linear model (such as logistic regression) can introduce covariates to adjust p , its ability to deal with the unobserved heterogeneity is limited.

In order to address the issue that binomial distribution cannot fit finite discrete data with over-dispersion, scholars propose a *beta-binomial* (Beta-Bino) distribution from the Bayesian viewpoint, where the success probability p is treated as a r.v. and a $\text{Beta}(\alpha, \beta)$ distribution is assigned as a prior of p , resulting in the pmf of $X \sim \text{Beta-Bino}(m, \alpha, \beta)$ as

$$\text{Beta-Bino}(x|m, \alpha, \beta) = \binom{m}{x} \frac{B(\alpha + x, \beta + m - x)}{B(\alpha, \beta)}, \quad x \in \mathbb{N}_{0:m}, \quad (1.1)$$

where $B(a, b)$ represents the beta function. This improvement significantly improves the robustness of the model by quantifying the prior uncertainty. The super parameters α and β can be interpreted as “the prior success/failure times”, and their ratio $\alpha/(\alpha + \beta)$ controls the prior mean, while $\alpha + \beta$ adjusts the prior intensity. The variance

$$\frac{m\alpha\beta(\alpha + \beta + m)}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

of the model explicitly contains the over-dispersion correction term, and its superiority has been verified in the analysis of gene expression heterogeneity and social science survey data. However, it still has two major limitations: first, the subjectivity of the choice of super-parameters, the setting of α and β depends on prior knowledge, which results the inference under small samples are sensitive. Second is the lack of complex heterogeneity modeling, when the data has a hierarchical structure (such as different subgroups have different p distribution), the standard model is difficult to capture population differences. Current research focuses on hierarchical Bayesian expansion: on the one hand, it introduces super prior distribution (such as α and β follow gamma distribution) to automatically learn the degree of dispersion. On the other hand, construct regression structure

$$\text{logit}(p_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n$$

to integrate covariates, and combine MCMC sampling to realize full Bayesian inference. These developments are promoting the application of beta-binomial model in frontier scenarios such as drug dose-response curve fitting and online user behavior prediction.

In fact, in the hierarchical model, the prior distribution of p does not necessarily have to be selected as beta distribution, but because beta distribution and binomial distribution

are conjugate, which makes beta-binomial distribution have explicit expression and easy to calculate. On the contrary, the prior of p can be regarded as any continuous distribution defined on the interval $[0, 1]$, such as simplex, logit-normal, PIG distribution and a series of distributions proposed by Xunjian Li in the article. The arbitrary choice of prior distribution makes the model fitting too subjective, and it is impossible to judge the true prior of the data in the actual model fitting. Therefore, in this paper, we use Bernstein polynomial and Weierstrass approximation theorem to propose a general accurate prior expression, which can include all the prior special cases mentioned above. We call such a general distribution as Bernstein-Binomial distribution.

The rest of the paper will be organized as follows. In Section 1.2, we will introduce Bernstein polynomial and Weierstrass approximation theorem. In Section 2, we define Bernstein-Binomial distribution for the first time and use MM algorithm, DA algorithm and other methods to conduct statistical inference. The quantile regression model of Bernstein-Binomial distribution will also be studied and in Section 2.4. Some numerical experiments are presented in Section 3, which can prove the reliability of our model and the accuracy of corresponding algorithms. In Section 4, we will apply this new model to real data analysis and compare it with some traditional models. The conclusion and discussion of the results are stated in Section 5. Finally, the technical details and derivation involved in the paper will be supplemented in the appendix.

1.2 Bernstein polynomials and Weierstrass approximation theorem

Given a non-negative integer K , Bernstein polynomials of order K is a set of $K + 1$ polynomials $\{B_{k,K}(\tau): k = 0, 1, \dots, K\}$, where the individual polynomial

$$B_{k,K}(\tau) \triangleq \binom{K}{k} \tau^k (1 - \tau)^{K-k} = \frac{1}{K+1} \cdot \text{Beta}(\tau|k+1, K-k+1), \quad \tau \in [0, 1], \quad (1.2)$$

is sometimes called as the k -th Bernstein basis function of order/degree K , here $\text{Beta}(\cdot|a, b)$ denotes the *probability density function* (pdf) of the $\text{Beta}(a, b)$ distribution.

Bernstein polynomials are particularly useful in approximation theory. Given a continuous function $f(\tau)$ on the interval $[0, 1]$, the K -order *Bernstein polynomial approximation*

(BPA) is defined as

$$\begin{aligned} B_K(\tau|f) &\triangleq \sum_{k=0}^K f(k/K) \cdot B_{k,K}(\tau) \stackrel{(1.2)}{=} \sum_{k=0}^K \frac{f(k/K)}{K+1} \cdot \text{Beta}(\tau|k+1, K-k+1) \\ &= \sum_{k=0}^K w_k \cdot \text{Beta}(\tau|k+1, K-k+1), \quad \tau \in [0, 1], \end{aligned} \quad (1.3)$$

where $w_k \triangleq f(k/K)/(K+1)$. The formula (1.3) indicates that the K -order BPA can be expressed as a linear combination of $K+1$ beta densities.

Theorem 1 (Weierstrass approximation theorem). For any continuous function $f(\tau)$ defined on $[0, 1]$, the K -order BPA $B_K(\tau|f)$ uniformly converges to $f(\tau)$; that is,

$$\lim_{K \rightarrow \infty} \sup_{\tau \in [0, 1]} |B_K(\tau|f) - f(\tau)| = 0. \quad \parallel$$

Consider a continuous r.v. $X^* \in [0, 1]$, if its pdf $f^*(\cdot)$ is a continuous function defined on $[0, 1]$, then Theorem 1 indicates that $f^*(\cdot)$ can be uniformly approximated by

$$\text{Bernstein}(\tau|K, \boldsymbol{\lambda}) \triangleq \sum_{k=0}^K \lambda_k \cdot \text{Beta}(\tau|k+1, K-k+1), \quad \tau \in [0, 1]. \quad (1.4)$$

The formula (1.4) defines the pdf of a so-called Bernstein($K, \boldsymbol{\lambda}$) distribution, where K is a known non-negative integer (or a predetermined constant), and $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \dots, \lambda_K)^\top$ is an unknown parameter vector of weights satisfying $\boldsymbol{\lambda} \in \mathbb{T}_{K+1}$, where

$$\mathbb{T}_n \triangleq \{\mathbf{x} = (x_1, \dots, x_n)^\top: 0 \leq x_i \leq 1, i = 1, \dots, n, \mathbf{x}^\top \mathbf{1}_n = 1\} \quad (1.5)$$

is the n -dimensional *closed* hyperplane, and $\mathbf{1}_n$ is the n -vector with each element being 1. For convenience, we define the n -dimensional *open* hyperplane as

$$\mathbb{T}_n^{\text{open}} \triangleq \{\mathbf{x} = (x_1, \dots, x_n)^\top: 0 < x_i < 1, i = 1, \dots, n, \mathbf{x}^\top \mathbf{1}_n = 1\}. \quad (1.6)$$

Because the Bernstein basis functions $\{B_{k,K}(\tau): k = 0, 1, \dots, K\}$ are linearly independent on the interval $[0, 1]$ and forms the basis of a $(K+1)$ -dimensional polynomial space. That is, any polynomial with a degree not exceeding K can be uniquely represented

as a linear combination of these basis functions. Therefore, the scaled beta distribution $\{\text{Beta}(\tau|k+1, K-k+1) \mid k = 0, 1, \dots, K\}$ is also linearly independent, ensuring the uniqueness and parameter identifiability of the Bernstein distribution as a finite discrete mixture of beta distributions.

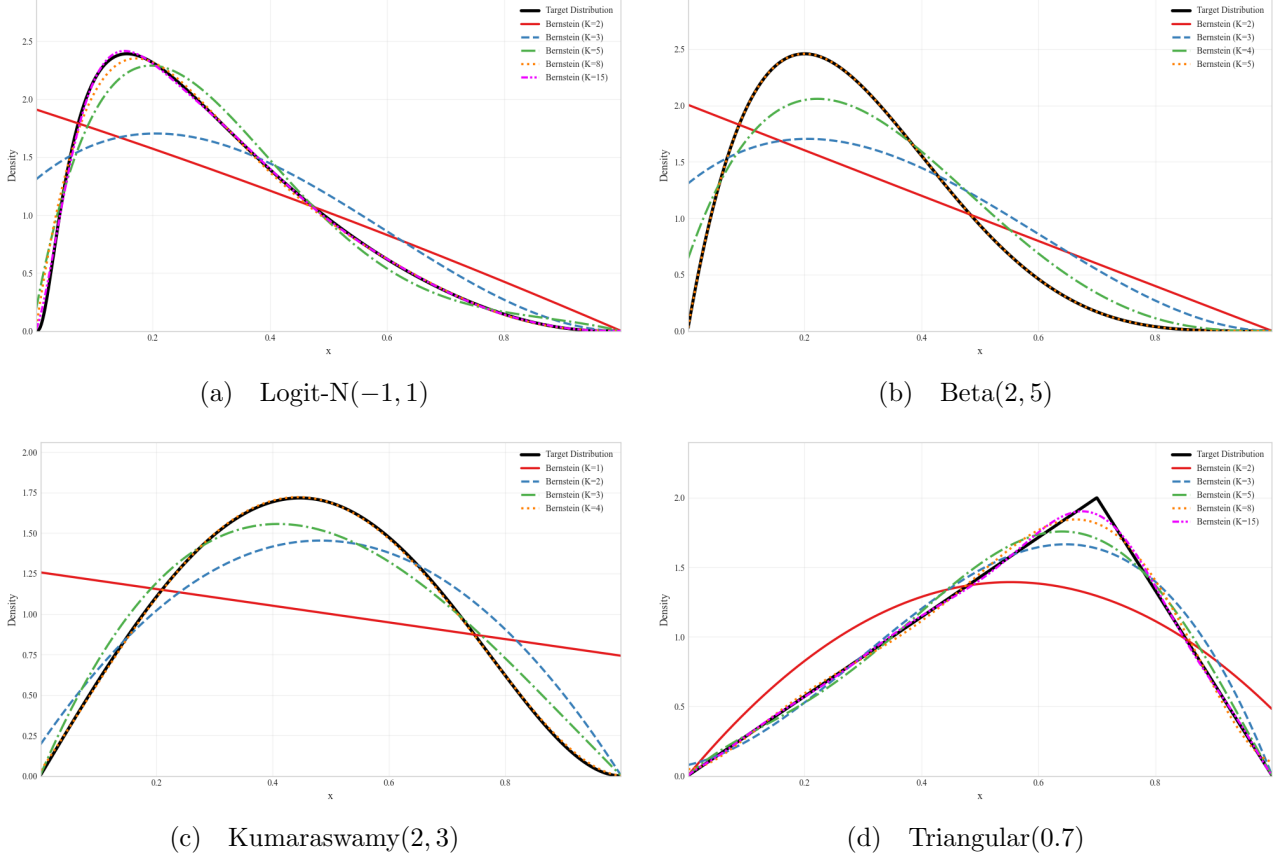


Figure 1: Bernstein approximation for various continuous distributions defined on $[0, 1]$.

Figure 1 shows the approximate effect of Bernstein distribution on some common continuous distributions defined on the interval $[0, 1]$. We can found that with the increase of K , the Bernstein distribution gradually approaches the target distribution, and finally achieves a completely approximate effect.

Based on the above theorems and conclusions, we can take Bernstein distribution as the general accurate prior distribution of the success probability p in binomial distribution, which can not only expand the shape constraint prior of the original beta-binomial distri-

bution, adapt to more complex and changeable real data, but also avoid the subjectivity and randomness of the selection of prior distribution, there is no need to preset the specific form.

2. Bernstein–Binomial distribution

2.1 Definition and basic properties

Let τ be a continuous r.v. taking values on the unit interval $[0, 1]$. If a discrete r.v. X has the following *mixture representation* (MR):

$$\tau \sim \text{Bernstein}(K, \boldsymbol{\lambda}) \quad \text{and} \quad X|\tau \sim \text{Binomial}(m, \tau), \quad (2.1)$$

then X is said to follow the *Bernstein–Binomial* (Bern-Bino) distribution, denoted by $X \sim \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$, where $\{m, K\}$ are two known constants and $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \dots, \lambda_K)^\top \in \mathbb{T}_{K+1}$ is an unknown weights vector. Its pmf is derived as

$$\text{Bern-Bino}(x|m, K, \boldsymbol{\lambda}) = \binom{m}{x} \sum_{k=0}^K a_k(x) \lambda_k = \binom{m}{x} [\boldsymbol{a}(x)]^\top \boldsymbol{\lambda}, \quad x \in \mathbb{N}_{0:m}. \quad (2.2)$$

where $\boldsymbol{a}(x) = (a_0(x), a_1(x), \dots, a_K(x))^\top$ and

$$a_k(x) = \frac{B(k+1+x, K-k+1+m-x)}{B(k+1, K-k+1)}, \quad k = 0, 1, \dots, K.$$

From the MR (2.1), on the one hand, we can obtain

$$\begin{aligned} E(X) &= \frac{m}{K+2} \sum_{k=0}^K (k+1) \lambda_k \quad \text{and} \\ \text{Var}(X) &= m \left\{ \frac{\sum_{k=0}^K (k+1) \lambda_k}{K+2} - \left[\frac{\sum_{k=0}^K (k+1) \lambda_k}{K+2} \right]^2 \right\} \\ &\quad + \frac{m(m-1)}{(K+2)^2(K+3)} \sum_{k=0}^K (k+1)(K-k+1) \lambda_k. \end{aligned}$$

On the other hand, we have the *stochastic representation* (SR) of $\tau \sim \text{Bernstein}(K, \boldsymbol{\lambda})$ as

$$\tau \stackrel{\text{d}}{=} Z_0 Y_0 + Z_1 Y_1 + \dots + Z_K Y_K = \boldsymbol{Z}^\top \boldsymbol{Y}, \quad (2.3)$$

where $\mathbf{Z} = (Z_0, Z_1, \dots, Z_K)^\top \sim \text{Multinomial}(1, \boldsymbol{\lambda})$, $\{Y_k\}_{k=0}^K \stackrel{\text{ind}}{\sim} \text{Beta}(k+1, K-k+1)$, and $\mathbf{Z} \perp \mathbf{Y} = (Y_0, Y_1, \dots, Y_K)^\top$. The SR (2.3) can be employed to generate random samples from the Bernstein($K, \boldsymbol{\lambda}$) distribution and in the *data augmentation* (DA) algorithm for Bayesian inferences in §3.

Remark 2.1 (Alternative expression of pmf for the Bern-Bino distribution). Based on (1.1), we can rewrite (2.2) as a mixture distribution of $(K+1)$ beta-binomial distributions:

$$\text{Bern-Bino}(x|m, K, \boldsymbol{\lambda}) \stackrel{(2.2)}{=} \sum_{k=0}^K \lambda_k \cdot \text{Beta-Bino}(x|m, k+1, K-k+1), \quad x \in \mathbb{N}_{0:m}. \quad (2.4)$$

Hence, similar to (2.3), we can obtain the SR of $X \sim \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ as

$$X \stackrel{d}{=} Z_0 V_0 + Z_1 V_1 + \dots + Z_K V_K = \mathbf{Z}^\top \mathbf{V}, \quad (2.5)$$

where $\mathbf{Z} = (Z_0, Z_1, \dots, Z_K)^\top \sim \text{Multinomial}(1, \boldsymbol{\lambda})$, $\{V_k\}_{k=0}^K \stackrel{\text{ind}}{\sim} \text{Beta-Bino}(m, k+1, K-k+1)$, and $\mathbf{Z} \perp \mathbf{V} = (V_0, V_1, \dots, V_K)^\top$. The SR (2.5) can be used to alternatively generate random samples from the Bern-Bino($m, K, \boldsymbol{\lambda}$) distribution parallel to the MR (2.1). $\parallel$

2.2 MLEs of parameters via an MM algorithm

Let $\{X_i\}_{i=1}^n \stackrel{\text{iid}}{\sim} \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ and $Y_{\text{obs}} = \{x_i\}_{i=1}^n$ denote the observations, then the log-likelihood function of $\boldsymbol{\lambda}$ is

$$\ell(\boldsymbol{\lambda}|Y_{\text{obs}}) \stackrel{(2.2)}{=} \sum_{i=1}^n \log \binom{m}{x_i} + \sum_{i=1}^n \log \left(\sum_{k=0}^K a_{ik} \lambda_k \right) = \text{constant} + \sum_{i=1}^n \log(\mathbf{a}_i^\top \boldsymbol{\lambda}), \quad (2.6)$$

where $\mathbf{a}_i = (a_{i0}, a_{i1}, \dots, a_{iK})^\top$,

$$a_{ik} = \frac{B(k+1+x_i, K-k+1+m-x_i)}{B(k+1, K-k+1)}, \quad i = 1, \dots, n, \quad k = 0, 1, \dots, K, \quad (2.7)$$

are positive constants free from $\boldsymbol{\lambda} \in \mathbb{T}_{K+1}$. The aim of this subsection is to calculate the MLEs of $\boldsymbol{\lambda}$ through an MM algorithm.

To this end, we first note the discrete version of Jensen's inequality

$$\varphi \left(\sum_{k=0}^K b_k z_k \right) \geq \sum_{k=0}^K b_k \varphi(z_k),$$

for any concave function $\varphi(\cdot)$, where $\mathbf{b} = (b_0, b_1, \dots, b_K)^\top \in \mathbb{T}_{K+1}$. For a fixed i , by setting $\varphi(\cdot) = \log(\cdot)$, we then have the following inequality

$$\log(\mathbf{a}_i^\top \boldsymbol{\lambda}) \geq \text{constant} + \sum_{k=0}^K b_{ik}^{(t)} \log(\lambda_k), \quad (2.8)$$

where the weights

$$b_{ik}^{(t)} \triangleq \frac{a_{ik} \lambda_k^{(t)}}{\sum_{l=0}^K a_{il} \lambda_l^{(t)}} = \frac{a_{ik} \lambda_k^{(t)}}{\mathbf{a}_i^\top \boldsymbol{\lambda}^{(t)}}, \quad k = 0, 1, \dots, K, \quad (2.9)$$

and $\boldsymbol{\lambda}^{(t)} = (\lambda_0^{(t)}, \lambda_1^{(t)}, \dots, \lambda_K^{(t)})^\top$ represents the t -th approximation of the MLE $\hat{\boldsymbol{\lambda}}$. By combining (2.6) with (2.8), we can construct a surrogate function as

$$Q(\boldsymbol{\lambda} | \boldsymbol{\lambda}^{(t)}) = \text{constant} + \sum_{i=1}^n \sum_{k=0}^K b_{ik}^{(t)} \log(\lambda_k) \propto \sum_{k=0}^K \left(\sum_{i=1}^n b_{ik}^{(t)} \right) \log(\lambda_k),$$

which satisfies $Q(\boldsymbol{\lambda} | \boldsymbol{\lambda}^{(t)}) \leq \ell(\boldsymbol{\lambda} | Y_{\text{obs}})$ and $Q(\boldsymbol{\lambda}^{(t)} | \boldsymbol{\lambda}^{(t)}) = \ell(\boldsymbol{\lambda}^{(t)} | Y_{\text{obs}})$.

Finally, by maximizing the $Q(\boldsymbol{\lambda} | \boldsymbol{\lambda}^{(t)})$ with respect to $\boldsymbol{\lambda}$, we obtain the MM iterations as

$$\lambda_k^{(t+1)} = \frac{\sum_{i=1}^n b_{ik}^{(t)}}{\sum_{l=0}^K \sum_{i=1}^n b_{il}^{(t)}} = \frac{\mathbf{1}_n^\top \mathbf{b}_k^{(t)}}{\mathbf{1}_n^\top \mathbf{B}^{(t)} \mathbf{1}_{K+1}}, \quad k = 0, 1, \dots, K, \quad (2.10)$$

where $\mathbf{B}^{(t)}$ is an $n \times (K+1)$ matrix defined by

$$\mathbf{B}^{(t)} \triangleq (b_{ik}^{(t)}) = (\mathbf{b}_0^{(t)}, \mathbf{b}_1^{(t)}, \dots, \mathbf{b}_K^{(t)}) \quad \text{with} \quad \mathbf{b}_k^{(t)} = \begin{pmatrix} b_{1k}^{(t)} \\ \vdots \\ b_{nk}^{(t)} \end{pmatrix}$$

being the k -column of $\mathbf{B}^{(t)}$, here $b_{ik}^{(t)}$ is given by (2.9). The MM iterations are repeated until

$$\left| \frac{\ell(\boldsymbol{\lambda}^{(t+1)} | Y_{\text{obs}}) - \ell(\boldsymbol{\lambda}^{(t)} | Y_{\text{obs}})}{\ell(\boldsymbol{\lambda}^{(t)} | Y_{\text{obs}})} \right| \leq \delta_0.$$

where δ_0 is a pre-specified small positive number.

2.3 Asymptotic normality of MLEs and CIs of parameters

Due to the existence of the constraint $\lambda_0 = 1 - \sum_{k=1}^K \lambda_k$ in $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \dots, \lambda_K)^\top$, we consider the parameter sub-vector $\boldsymbol{\lambda}_{-0} = (\lambda_1, \dots, \lambda_K)^\top$, where $\lambda_k \in (0, 1)$ (we do not consider the

cases of parameter components with $\lambda_k = 0$ or $\lambda_k = 1$, indicating that the k -th weight is zero or one). One of the key questions in large-sample statistical inferences is to verify the asymptotic normality of MLEs; i.e.,

$$\sqrt{n} \left(\hat{\boldsymbol{\lambda}}_{-0,n} - \boldsymbol{\lambda}_{-0,\text{true}} \right) \xrightarrow{D} N(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{as } n \rightarrow \infty,$$

where $\hat{\boldsymbol{\lambda}}_{-0,n}$ denotes the MLE of $\boldsymbol{\lambda}_{-0}$ for emphasizing the dependency of $\hat{\boldsymbol{\lambda}}_{-0}$ on the sample size n , $\boldsymbol{\lambda}_{-0,\text{true}}$ is the true value of $\boldsymbol{\lambda}_{-0}$ and $\boldsymbol{\Sigma}$ is the variance-covariance matrix of the asymptotic normal distribution. The asymptotic normality depends on a series of assumptions known as regularization or Cramér–Rao conditions (Lehmann & Casella 1998 and Lehmann 1999), which are presented in Theorem 2 of Appendix A.1. By verifying the Cramér–Rao conditions for Bern–Bino distribution (see Appendix A.2 for more details), we obtain the asymptotic normality of $\hat{\boldsymbol{\lambda}}_{-0,n}$ as

$$\sqrt{n} \left(\hat{\boldsymbol{\lambda}}_{-0,n} - \boldsymbol{\lambda}_{-0,\text{true}} \right) \xrightarrow{D} N \left(\mathbf{0}, \mathbf{J}^{-1}(\boldsymbol{\lambda}_{-0,\text{true}}) \right) \quad \text{as } n \rightarrow \infty, \quad (2.11)$$

where $\mathbf{J}(\boldsymbol{\lambda}_{-0,\text{true}})$ is the Fisher information matrix. After obtaining (2.11), by using the delta method, we obtain the asymptotic distribution of $\hat{\lambda}_0 \triangleq g(\hat{\boldsymbol{\lambda}}_{-0}) = 1 - \sum_{k=1}^K \hat{\lambda}_k$ as

$$\sqrt{n}(\hat{\lambda}_{0,n} - \lambda_{0,\text{true}}) \xrightarrow{D} N \left(0, [\nabla g(\boldsymbol{\lambda}_{-0,\text{true}})]^\top \mathbf{J}^{-1}(\boldsymbol{\lambda}_{-0,\text{true}}) \nabla g(\boldsymbol{\lambda}_{-0,\text{true}}) \right).$$

In practice, we adopt the sandwich covariance matrix estimation to replace $\mathbf{J}^{-1}(\boldsymbol{\lambda}_{-0,\text{true}})$, which can be computed as

$$\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{H}}^{-1} \hat{\mathbf{S}} \hat{\mathbf{H}}^{-1},$$

where $\hat{\mathbf{S}}$ is the outer product of scores and $\hat{\mathbf{H}}$ is the observed information matrix

$$\begin{aligned} \hat{\mathbf{S}} &= \frac{1}{n} \sum_{i=1}^n \mathbf{s}_i \mathbf{s}_i^\top \quad \text{with} \quad \mathbf{s}_i = \nabla \log p(x_i | \boldsymbol{\lambda}_{-0}) \Big|_{\boldsymbol{\lambda}_{-0} = \hat{\boldsymbol{\lambda}}_{-0,n}} \quad \text{and} \\ \hat{\mathbf{H}} &= -\nabla^2 \log \ell(\boldsymbol{\lambda}_{-0} | Y_{\text{obs}}) \Big|_{\boldsymbol{\lambda}_{-0} = \hat{\boldsymbol{\lambda}}_{-0,n}}. \end{aligned}$$

Finally, we construct the $100(1 - \alpha)\%$ asymptotic CI of λ_k as $\hat{\lambda}_{k,n} \pm z_{\alpha/2} \cdot \hat{\text{se}}_k$ with standard error $\hat{\text{se}}_k = \sqrt{\hat{\boldsymbol{\Sigma}}_{kk}}$ for $k = 1, \dots, K$ and $\hat{\text{se}}_0 = \sqrt{[\nabla g(\hat{\boldsymbol{\lambda}}_{-0})]^\top \hat{\boldsymbol{\Sigma}} \nabla g(\hat{\boldsymbol{\lambda}}_{-0})}$.

2.4 Bootstrap confidence intervals of λ

The large-sample method to calculate CIs of parameters has two limitations: (1) The asymptotic CI is not reliable for small to moderate sample sizes; and (2) even though for a large sample size, the range of CI may be beyond its bounds when the true value is close to the bounds. To overcome the two difficulties, this subsection presents the bootstrap method to construct the *bootstrap confidence interval* (BCI) of $\vartheta \triangleq h(\lambda)$, where $h(\cdot)$ is an arbitrary function of the parameter vector λ in the Bern-Bino(m, K, λ) distribution.

Based on the observations $Y_{\text{obs}} = \{x_i\}_{i=1}^n$, we can first compute the MLEs $\hat{\lambda}$ of λ through the MM algorithm (2.10). Then, we generate a bootstrap sample $\{X_i^* = x_i^*\}_{i=1}^n \stackrel{\text{iid}}{\sim} \text{Bern-Bino}(m, K, \hat{\lambda})$ and compute one bootstrap replication $\hat{\vartheta}^* = h(\hat{\lambda}^*)$. Independently repeating this process G times, we obtain G bootstrap replications $\{\hat{\vartheta}^*(g)\}_{g=1}^G$. Hence, a $100(1 - \alpha)\%$ BCI of ϑ is $[\hat{\vartheta}_L^*, \hat{\vartheta}_U^*]$, where $\hat{\vartheta}_L^*$ and $\hat{\vartheta}_U^*$ are the $(\alpha/2)G$ -th and the $(1 - \alpha/2)G$ -th order statistics of $\{\hat{\vartheta}^*(g)\}_{g=1}^G$.

2.5 Regression model

Assume that $\{X_i\}_{i=1}^n \stackrel{\text{ind}}{\sim} \text{Bern-Bino}(m, K, \lambda_i)$ and the observed data be $Y_{\text{obs}} = \{x_i, \mathbf{z}_{(i)}\}$, where $\mathbf{z}_{(i)} = (z_{i1}, \dots, z_{ip})^\top$ is the covariate vector of the i -th subject. We consider the following regression model

$$\lambda_{ik} = \frac{\exp(\eta_{ik})}{\sum_{l=0}^K \exp(\eta_{il})}, \quad i = 1, \dots, n, \quad \text{where}$$

$$\eta_{ik} = \phi_k + \gamma_k \cdot \mathbf{z}_{(i)}^\top \boldsymbol{\beta}, \quad k = 0, 1, \dots, K \quad \text{with} \quad \phi_0 = 0, \gamma_0 = 1.$$

The log-likelihood function of $\boldsymbol{\theta} = (\boldsymbol{\phi}^\top, \boldsymbol{\gamma}^\top, \boldsymbol{\beta}^\top)^\top$ is given by

$$\ell(\boldsymbol{\theta} | Y_{\text{obs}}) = \sum_{i=1}^n \log \binom{m}{x_i} + \sum_{i=1}^n \log \left(\sum_{k=0}^K a_{ik} \lambda_{ik} \right).$$

By calculating the score vector $\mathbf{s}(\boldsymbol{\theta})$ and Hessian matrix $\mathbf{H}(\boldsymbol{\theta})$, we adopt the Newton-Raphson algorithm to compute the MLE $\hat{\boldsymbol{\theta}}$ of parameter $\boldsymbol{\theta}$ as

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \mathbf{H}^{-1}(\boldsymbol{\theta}^{(t)}) \mathbf{s}(\boldsymbol{\theta}^{(t)}).$$

3. Bayesian inferences

Let $\{X_i\}_{i=1}^n \stackrel{\text{iid}}{\sim} \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ and the observed data $Y_{\text{obs}} = \{x_i\}_{i=1}^n$. We perform Bayesian inferences on parameters $\boldsymbol{\lambda} = (\lambda_0, \lambda_1, \dots, \lambda_K)^\top \in \mathbb{T}_{K+1}$; i.e., we will calculate the posterior mode via an MM algorithm and generate posterior samples via a DA algorithm. Finally, the posterior mean and Bayesian credible interval can be obtained based on posterior samples.

Consider the prior $\boldsymbol{\lambda} \sim \text{Dirichlet}_{K+1}(\boldsymbol{\alpha})$ with joint pdf

$$\text{Dirichlet}_{K+1}(\boldsymbol{\lambda}|\boldsymbol{\alpha}) = \frac{\Gamma(\sum_{k=0}^K \alpha_k)}{\prod_{k=0}^K \Gamma(\alpha_k)} \prod_{k=0}^K \lambda_k^{\alpha_k-1}, \quad \boldsymbol{\lambda} \in \mathbb{T}_{K+1}, \quad (3.1)$$

where $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \dots, \alpha_K)^\top$ is a known vector. Then, we obtain the posterior distribution

$$p(\boldsymbol{\lambda}|Y_{\text{obs}}) \propto \mathcal{L}(\boldsymbol{\lambda}|Y_{\text{obs}}) \cdot \text{Dirichlet}_{K+1}(\boldsymbol{\lambda}|\boldsymbol{\alpha}), \quad (3.2)$$

where $\mathcal{L}(\boldsymbol{\lambda}|Y_{\text{obs}})$ is the likelihood function of $\boldsymbol{\lambda}$. Find the mode of posterior distribution is equivalent to optimizing $\log p(\boldsymbol{\lambda}|Y_{\text{obs}})$ to obtain the optimized value $\tilde{\boldsymbol{\lambda}}$.

3.1 The MM algorithm to calculate the posterior mode of $\boldsymbol{\lambda}$

From (3.2) with the prior (3.1), we easily obtain

$$\log p(\boldsymbol{\lambda}|Y_{\text{obs}}) = \text{constant} + \sum_{i=1}^n \log \left(\sum_{k=0}^K a_{ik} \lambda_k \right) + \sum_{k=0}^K (\alpha_k - 1) \log(\lambda_k).$$

Similar to §2.2, we employ Jensen's inequality again to construct the surrogate function

$$Q(\boldsymbol{\lambda}|\boldsymbol{\lambda}^{(t)}) = \text{constant} + \sum_{k=0}^K \left(\sum_{i=1}^n b_{ik}^{(t)} + \alpha_k - 1 \right) \log(\lambda_k),$$

where $b_{ik}^{(t)}$ is defined by (2.9). Similar to (2.10), we obtain the MM iterations as

$$\lambda_k^{(t+1)} = \frac{\sum_{i=1}^n b_{ik}^{(t)} + \alpha_k - 1}{\sum_{l=0}^K (\sum_{i=1}^n b_{il}^{(t)} + \alpha_l - 1)}, \quad k = 0, 1, \dots, K.$$

3.2 The DA algorithm to generate posterior samples

From (2.3), we know that $\mathbf{Z} = (Z_0, Z_1, \dots, Z_K)^\top \sim \text{Multinomial}(1, \boldsymbol{\lambda})$ is equivalent to a finite discrete r.v. W defined on $\mathbb{N}_{0:K}$ with corresponding probabilities $\boldsymbol{\lambda}$, where $W = k$ means $Z_k = 1$ is sampled from the k -th category. Therefore, we introduce the missing data $Y_{\text{mis}} = \{w_i\}_{i=1}^n$, where the latent variable $W_i \stackrel{\text{iid}}{\sim} \text{FDiscrete}(\mathbb{N}_{0:K}, \boldsymbol{\lambda})$ with $p(W_i = k) = \lambda_k$ for $k = 0, 1, \dots, K$. Therefore, the complete data can be expressed as $Y_{\text{com}} = \{Y_{\text{obs}}, Y_{\text{mis}}\}$.

In order to make a full Bayesian inference on the parameter vector $\boldsymbol{\lambda}$, we could employ the DA algorithm of Tanner & Wong (1987), where the **I-step** is to generate the missing data $\{W_i = w_i\}_{i=1}^n$ from the conditional predictive distribution

$$\begin{aligned} p(W_i = k | x_i, \boldsymbol{\lambda}) &\propto p(x_i | W_i = k, \boldsymbol{\lambda}) \cdot p(W_i = k | \boldsymbol{\lambda}) \\ &= \lambda_k \cdot \binom{m}{x_i} \frac{B(x_i + k + 1, m - x_i + K - k + 1)}{B(k + 1, K - k + 1)} \\ &= \lambda_k \cdot \text{Beta-Bino}(x_i | m, k + 1, K - k + 1) \triangleq q_{ik}; \end{aligned}$$

i.e., $W_i | (x_i, \boldsymbol{\lambda}) \sim \text{FDiscrete}(\mathbb{N}_{0:K}, \mathbf{p}_i)$, where $\mathbf{p}_i \triangleq (p_{i0}, p_{i1}, \dots, p_{iK})^\top$ with $p_{ik} \propto q_{ik}$. The **P-step** is draw $\boldsymbol{\lambda}$ from the complete-data posterior distribution

$$p(\boldsymbol{\lambda} | Y_{\text{com}}) = p(\boldsymbol{\lambda} | \mathbf{W}) = \pi(\boldsymbol{\lambda} | \boldsymbol{\alpha}) \cdot p(\mathbf{W} | \boldsymbol{\lambda}) \propto \prod_{k=0}^K \lambda_k^{\alpha_k + N_k - 1} \sim \text{Dirichlet}_{K+1}(\boldsymbol{\alpha} + \mathbf{N}),$$

where $\mathbf{W} = (W_1, \dots, W_n)^\top$ and $\mathbf{N} = (N_0, N_1, \dots, N_K)^\top$ with $N_k = \sum_{i=1}^n I(W_i = k)$. Thus, we summarize the DA algorithm for calculating the posterior mean and Bayesian CIs of λ_k s as follows:

Algorithm 1: Posterior mean and Bayesian CI via the DA algorithm

Input : The initial parameter vector $\boldsymbol{\lambda}^{(0)}$, latent variable $\mathbf{W}^{(0)}$, the observed data $Y_{\text{obs}} = \{x_i\}_{i=1}^n$ and the sampling number G .

Output: Posterior mean $\bar{\lambda}_k$ and Bayesian CI $[\lambda_k^L, \lambda_k^U]$ for $k = 0, 1, \dots, K$.

- 1 Set initial vectors $\boldsymbol{\lambda}^{(0)}$ and $\mathbf{W}^{(0)}$;
 - 2 **for** $g = 0$ **to** G **do**
 - 3 **l-step:** Draw latent variables $\mathbf{W}^{(g+1)}$ from $W_i | \boldsymbol{\lambda}^{(g)} \sim \text{FDiscrete}(\mathbb{N}_{0:K}, \mathbf{p}_i^{(g)})$,
 $i = 1, \dots, n$, where $\mathbf{p}_i^{(g)} = (p_{i0}^{(g)}, p_{i1}^{(g)}, \dots, p_{iK}^{(g)})^\top$ with $p_{ik}^{(g)} \propto q_{ik}^{(g)}$.
 - 4 **P-step:** Draw parameter vector $\boldsymbol{\lambda}^{(g+1)}$ from
 $\boldsymbol{\lambda} | \mathbf{W}^{(g+1)} \sim \text{Dirichlet}_{K+1}(\boldsymbol{\alpha} + \mathbf{N}^{(g+1)})$, where $\mathbf{N}^{(g)} = (N_0^{(g)}, N_1^{(g)}, \dots, N_K^{(g)})^\top$
with $N_k^{(g)} = \sum_{i=1}^n I(W_i^{(g)} = k)$.
 - 5 **end**
 - 6 **Collection of posterior samples:** Abandon the results of the previous B
iterations and save the remaining samples $\{\boldsymbol{\lambda}^{(g)}\}_{g=B+1}^G$.
 - 7 **Compute the posterior mean:** $\bar{\lambda}_k \triangleq [1/(G-B)] \sum_{g=B+1}^G \lambda_k^{(g)}$, $k = 0, 1, \dots, K$.
 - 8 **Compute the Bayesian CI:** $[\lambda_k^L, \lambda_k^U] = [\lambda_k^{(\alpha/2)}, \lambda_k^{(1-\alpha/2)}]$, $k = 0, 1, \dots, K$, where
 $\lambda_k^{(q)}$ is the q -quantile of the posterior sample.
-

4. Simulation studies

4.1 Accuracy of MLEs of parameters and reliability of CIs

In §2, we proposed the Bern-Bino($m, K, \boldsymbol{\lambda}$) distribution and calculated the MLEs of $\boldsymbol{\lambda}$ via an MM algorithm. In addition, the CIs of parameters can be obtained by the sandwich method for large sample sizes or by the bootstrap method for the small to moderate sample sizes. To assess the performance of the proposed algorithm and methods, we conduct some numerical experiments by employing the *mean square error* (MSE) as the criterion to evaluate the accuracy of the MLE of each parameter and apply the *coverage rate* (CR) and *interval width* (I-Width) to confirm the reliability of CIs. Consider the sample sizes to be $n = 200(200)$ 1,000 and the number of sampling replications to be $R = 10,000$ in our simulations. We conduct some numerical experiments for the following two cases:

Case (A₁): Independently generating $\{X_i^{(r)}\}_{i=1}^n \stackrel{\text{ind}}{\sim} \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ with the true values being set as $m = 8$, $K = 2$, $\boldsymbol{\lambda} = (0.2, 0.5, 0.3)^\top$;

Case (A₂): Independently generating $\{X_i^{(r)}\}_{i=1}^n \stackrel{\text{ind}}{\sim} \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ with the true values being set as $m = 15$, $K = 3$, $\boldsymbol{\lambda} = (0.15, 0.3, 0.35, 0.2)^\top$;

Given a combination of $\{n, r, m, K, \boldsymbol{\lambda}\}$ with $r = 1, \dots, R$, for each case, let $Y_{\text{obs}}^{(r)} = \{x_i^{(r)}\}_{i=1}^n$ denote the r -th generated observations, where $x_i^{(r)}$ is a realization of $X_i^{(r)}$. We can compute the MLEs of the parameter vector $\boldsymbol{\lambda}$ via the MM algorithm (2.10), denoted by $\hat{\boldsymbol{\lambda}}^{(r)}$. Next, we calculate the *average MLE* (A-MLE), MSE based on the R repetitions. Finally, we calculate the CR and I-Width of CIs, which are obtained by the sandwich method. All these results are presented in Table 1.

Table 1. Comparisons of A-MLE, MSE, CR and I-Width based on $R = 10,000$ sampling replications in Bern-Bino distributions for Cases (A₁)–(A₂) with various sample sizes

Case	Parameter	Criterion	Sample size				
			200	400	600	800	1000
(A ₁)	λ_0	A-MLE	0.2021	0.2011	0.2005	0.2007	0.2011
		MSE	0.0065	0.0032	0.0021	0.0016	0.0013
		CR	0.9445	0.9525	0.9497	0.9475	0.9462
		I-Width	0.3156	0.2237	0.1827	0.1583	0.1416
	λ_1	A-MLE	0.4966	0.4980	0.4991	0.4985	0.4980
		MSE	0.0203	0.0102	0.0067	0.0051	0.0042
		CR	0.9475	0.9506	0.9503	0.9492	0.9458
		I-Width	0.5606	0.3973	0.3245	0.2811	0.2515
	λ_2	A-MLE	0.3013	0.3009	0.3004	0.3009	0.3008
		MSE	0.0075	0.0038	0.0025	0.0019	0.0016
		CR	0.9476	0.9532	0.9477	0.9502	0.9465
		I-Width	0.3403	0.2412	0.1970	0.1707	0.1527
(A ₂)	λ_0	A-MLE	0.1493	0.1488	0.1498	0.1492	0.1496
		MSE	0.0054	0.0029	0.0020	0.0015	0.0012
		CR	0.9168	0.9408	0.9497	0.9524	0.9499
		I-Width	0.3492	0.2287	0.1805	0.1570	0.1396
	λ_1	A-MLE	0.3024	0.3025	0.3006	0.3024	0.3016
		MSE	0.0315	0.0186	0.0127	0.0093	0.0077
		CR	0.8789	0.9444	0.9578	0.9586	0.9520
		I-Width	0.8380	0.5784	0.4552	0.3970	0.3525
	λ_2	A-MLE	0.3490	0.3480	0.3504	0.3474	0.3487
		MSE	0.0338	0.0199	0.0135	0.0097	0.0081
		CR	0.8833	0.9479	0.9522	0.9581	0.9534
		I-Width	0.8181	0.5934	0.4674	0.4075	0.3620
	λ_3	A-MLE	0.1993	0.2007	0.1993	0.2010	0.2001
		MSE	0.0066	0.0036	0.0024	0.0018	0.0015
		CR	0.9219	0.9448	0.9486	0.9555	0.9486
		I-Width	0.3664	0.2471	0.1957	0.1704	0.1517

The convergence accuracy of A-MLE is 10^{-8} .

From the results in Table 1, we can found that the proposed MM algorithm is very accurate in estimating the parameters. As the sample size increases, the MSE and I-Width

gradually decreases, which makes the estimation more robust. However, in the case of small samples, the asymptotic normality of the parameters is poor, so the problem of insufficient interval coverage rate may occur when the number of parameters is large (like Case (A₂) with sample size $n = 200$). In such case, we can select the bootstrap method as an alternative to obtain the accurate CI.

4.2 Comparison of Bernstein prior generalization

In order to systematically evaluate the performance of the our new distribution constructed based on Bernstein prior in complex data scenarios, a series of numerical simulation experiments are designed and carried out in this section. The purpose of this study is to compare and analyze the advantages and limitations of Bern-Bino distribution and existing mainstream distribution in terms of fitting accuracy and robustness by simulating complex data from different real prior distributions. Specifically, we will generate the success probability parameter p of binomial distribution from several common distributions with different morphological characteristics, and then generate the corresponding binomial discrete observation samples, so as to test the ability of the new model in capturing the potential a prior structural complexity and mitigating the deviation of model setting.

To comprehensively and objectively quantify the fitting effect of each model, we use multiple evaluation criteria for comprehensive comparison: including the log-likelihood function value of the model, which is used to directly measure the fitting degree of the model to the data. *Akaike information criterion* (AIC), which introduces penalty term on the basis of likelihood to balance model complexity and goodness of fit and avoid over fitting. And χ^2 goodness of fit test, the absolute fitting effect is evaluated by comparing the observation frequency with the model prediction frequency, where the corresponding p -value is provided as the judgment basis.

In the experimental design, the sample size, the number of trials in binomial distribution, the number of sampling replications are respectively set to be $n = 500$, $m = 30$ and $R = 1,000$. We selected four representative distributions as the real prior of generating binomial data success probability p to simulate the possible diversity of prior information in the real

world: beta distribution, logit-normal (Logit-N) distribution, and two complex mixed models called Logistic-Stdlogistic (LS) distribution and Kumaraswamy (KM) distribution. For a given $\{n, r, \boldsymbol{\theta}\}$ with $r = 1, \dots, R$ and $\boldsymbol{\theta} = (\alpha_1, \beta_1, \mu_2, \sigma_2, \mu_3, \sigma_3, \alpha_4, \beta_4)^\top$, we independently generate the prior $P_i^{(r)} = p_i^{(r)}$ from the mixed distribution:

$$\omega_1 \cdot \text{Beta}(\alpha_1, \beta_1) + \omega_2 \cdot \text{Logit-N}(\mu_2, \sigma_2) + \omega_3 \cdot \text{LS}(\mu_3, \sigma_3) + \omega_4 \cdot \text{KM}(\alpha_4, \beta_4),$$

where $\omega_s \geq 0$ for $s = 1, \dots, 4$ with $\sum_{s=1}^4 \omega_s = 1$. We consider the following situations for selecting $\boldsymbol{\omega} = (\omega_1, \dots, \omega_4)^\top$ and parameters in four prior distribution:

Case (B₁): Uniform four-component mixed distribution: $\boldsymbol{\omega} = 0.25 \times \mathbf{1}_4$ with parameters $\boldsymbol{\theta} = (1, 3, -1, 0.5, 2, 5, 1, 2)^\top$;

Case (B₂): Dominant one-component mixed distribution: One $\omega_s = 0.4$, while others $\omega_{s'} = 0.2$ for $s' \neq s$ with parameters $\boldsymbol{\theta} = (1, 3, -1, 0.5, 2, 5, 1, 2)^\top$;

Case (B₃): A complex mixed distribution with $\boldsymbol{\omega} = 0.25 \times \mathbf{1}_4$ and parameter vector $\boldsymbol{\theta} = (1, 0.5, 0, 0.25, 2, 5, 1, 5)^\top$.

By sampling from the mixed distribution, based on the generated prior $\{p_1^{(r)}, \dots, p_n^{(r)}\}$, we further generate the corresponding binomial distribution sample $\{x_1^{(r)}, \dots, x_n^{(r)}\}$. Then we fit the corresponding binomial mixture distribution and obtain their log-likelihoods, AICs, p -values for χ^2 goodness of fit test, which is proposed by Karl Pearson (1900) to evaluate whether the deviation between the observation frequency (O_i) and the expected frequency (E_i) under the theoretical distribution is caused by random sampling error. The test statistics are the sum of squares of standardized residuals:

$$\chi_0^2 = \sum_{j=0}^m \frac{(O_i - E_i)^2}{E_i},$$

and the corresponding p -value can be computed as $p = \Pr(\chi^2 > \chi_0^2 \mid \nu)$ with degree of freedom ν .

We obtain the corresponding five ranks $\{\text{Rank}_{1,A}^{(r)}, \dots, \text{Rank}_{5,A}^{(r)}\}$, where $\text{Rank}_{s,A}^{(r)}$ represents the rank of $\text{AIC}_s^{(r)}$ in $\{\text{AIC}_1^{(r)}, \dots, \text{AIC}_5^{(r)}\}$. And $\{\text{Rank}_{1,p}^{(r)}, \dots, \text{Rank}_{5,p}^{(r)}\}$ represents the rank of

$\{p\text{-value}_1^{(r)}, \dots, p\text{-value}_5^{(r)}\}$. Finally, the average log-likelihoods (denoted by **Log-likelihood**), AICs (denoted by **AIC**), p -value (**p-value**) and two ranks (**Rank_A**, **Rank_p**) are calculated by

$$\begin{aligned} \text{Log-likelihood}_s &= \frac{1}{R} \sum_{r=1}^R \text{Log-likelihood}_s^{(r)}, \quad \text{AIC}_s = \frac{1}{R} \sum_{r=1}^R \text{AIC}_s^{(r)} \\ \text{p-value}_s &= \frac{1}{R} \sum_{r=1}^R \text{p-value}_s^{(r)} \quad \text{and} \quad \text{Rank}_{s,t} = \frac{1}{R} \sum_{r=1}^R \text{Rank}_{s,t}^{(r)}, \quad s = 1, \dots, 5, \quad t = A, p. \end{aligned}$$

Table (2) presents the **Log-likelihoods**, **AICs**, **Rank_As**, **p-values** and **Rank_ps** of five fitted distributions. From the table, we can found that, in Cases (B₁) and (B₂), Bernstein prior has excellent fitting effect on the simulated data. The log-likelihood value of Bernstein prior fitting is the largest, which means Bern-Bino distribution has the strongest ability to interpret the data. And the AIC value is the smallest among all models, indicating that it achieves a good balance between fitting accuracy and complexity. At the same time, the p -value of chi square test of Bern-Bino distribution is also the largest, therefore, there is no significant difference between the observed data and the distribution we fit. These results show that in the common mixed prior scenario, the performance of the Bern-Bino distribution is not inferior to that of traditional models, but rather it has more advantages in fitting robustness and generalization ability. And in the complex mixed situation of Case (B₃), severe fitting failures occurred in various traditional simple distributions, the p -values of their χ^2 tests are almost zero. This indicates that these distributions are no longer effectively capable the complex prior heterogeneous structures. In contrast, the Bern-Bino distribution still maintains excellent fitting performance. Its log-likelihood value is significantly higher, the AIC is significantly lower, and the p -value reaches 0.4863, which is much higher than that of other distributions, which fully demonstrates that Bern-Bino distribution can still stably and accurately depict the true form under complex structure. In conclusion, we think Bern-Bino distribution perform excellent with outstanding flexibility and robustness.

Table 2. Comparison of Log-likelihood, AIC, Rank_A, p-value and Rank_p based on $R = 1,000$ sampling replications for five distributions in Cases (B₁)–(B₃).

Case	Prior	Log-likelihood	AIC	Rank _A	p-value	Rank _p
(B ₁)	Beta	−1559.059	3122.119	3.2401	0.1199	3.0947
	Logit-N	−1565.753	3135.507	1.0322	0.0318	1.3030
	LS	−1561.722	3127.443	2.3897	0.0918	2.5592
	KM	−1558.132	3120.265	4.2008	0.1248	3.0507
	Bernstein	−1551.174	3118.432	4.1373	0.5841	4.9924
(B ₂) with $\omega = (0.4, 0.2, 0.2, 0.2)^\top$	Beta	−1558.663	3121.326	3.4688	0.1768	3.2202
	Logit-N	−1566.051	3136.102	1.1113	0.0406	1.2893
	LS	−1563.694	3131.387	2.0374	0.0837	2.2088
	KM	−1557.889	3119.777	4.4555	0.1852	3.2945
	Bernstein	−1552.819	3119.563	3.9271	0.5620	4.9872
(B ₂) with $\omega = (0.2, 0.4, 0.2, 0.2)^\top$	Beta	−1534.893	3073.787	3.2841	0.0923	2.8423
	Logit-N	−1539.508	3083.015	1.1463	0.0433	1.7372
	LS	−1534.770	3073.540	3.4252	0.1597	3.2098
	KM	−1534.540	3073.080	3.6070	0.0814	2.2315
	Bernstein	−1527.199	3071.843	3.5374	0.5902	4.9792
(B ₂) with $\omega = (0.2, 0.2, 0.4, 0.2)^\top$	Beta	−1553.071	3110.141	2.9413	0.0362	3.0791
	Logit-N	−1560.936	3125.873	1.0019	0.0074	1.6117
	LS	−1555.392	3114.784	2.3627	0.0208	2.0559
	KM	−1550.760	3105.519	4.1136	0.0473	3.2571
	Bernstein	−1540.049	3099.329	4.5805	0.5029	4.9962
(B ₂) with $\omega = (0.2, 0.2, 0.2, 0.4)^\top$	Beta	−1580.142	3164.284	3.4138	0.1786	3.1506
	Logit-N	−1586.090	3176.180	1.0393	0.0543	1.1809
	LS	−1582.358	3168.717	2.5563	0.1518	2.7334
	KM	−1579.774	3163.549	4.0417	0.1777	2.9455
	Bernstein	−1574.369	3162.118	3.9489	0.5741	4.9896
(B ₃)	Beta	−1702.888	3409.775	2.8097	0.0005	2.5005
	Logit-N	−1714.498	3432.997	1.0000	0.0005	2.5005
	LS	−1697.513	3399.026	4.0000	0.0005	2.5024
	KM	−1702.944	3409.888	2.1903	0.0005	2.5005
	Bernstein	−1653.137	3330.307	5.0000	0.4863	4.9962

In addition, we also generate sample and fit distributions under large samples to explore the performance of Bernstein binomial distribution for Cases (B₁) and (B₃), we set the

sample size $n = 5,000$ to generate mixed sample and use the above five distributions to perform the fitting and comparison.

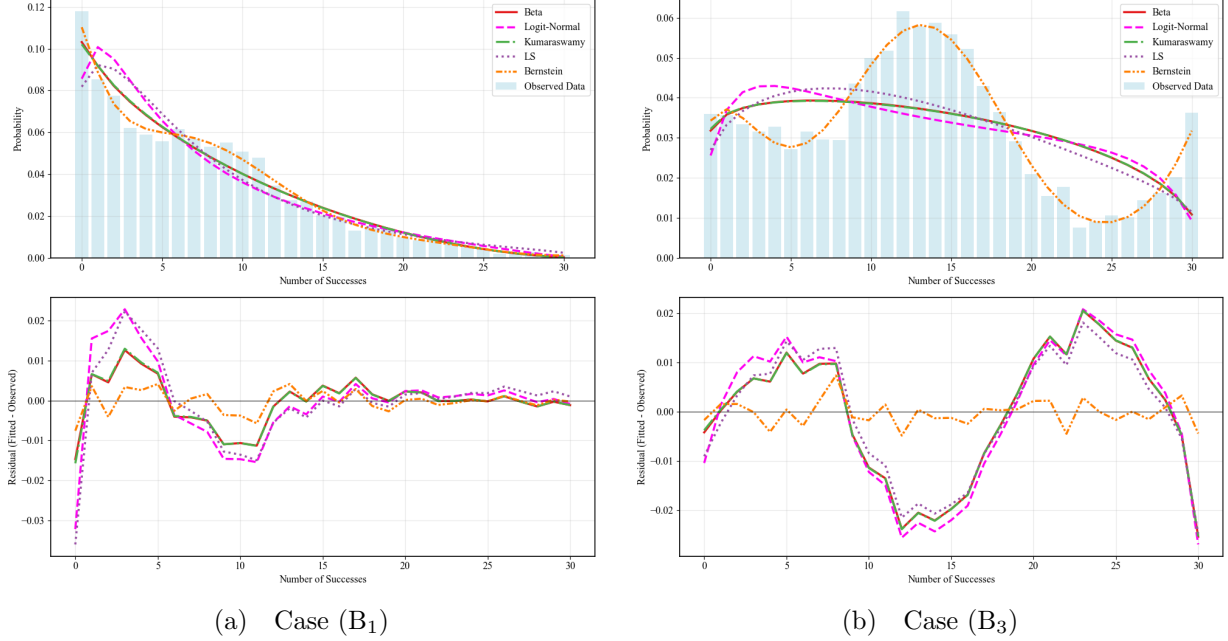


Figure 2: Fitting effect of five distributions and the corresponding residual plots.

Figure 2 shows the histogram of the generated samples together with the fitting curve of each distribution, and the residuals of the model fitting, which visually shows the proximity of each model to the empirical distribution. Sub-figure (a) and (b) in Figure 3 uses the Log-likelihood value, AIC, BIC, p -value of χ^2 goodness test, *Mean absolute error* (MAE) as evaluation criteria to evaluate the fitting effect of five models, and the comprehensive performance of each distribution are clearly displayed by radar chart. We can see that Bern-Bino has the largest area in the radar chart, which performs best. Sub-figure (c) and (d) represents the Bern-Bino distribution gradually approximates the real distribution with the increase of K , showing the flexibility and convergence.

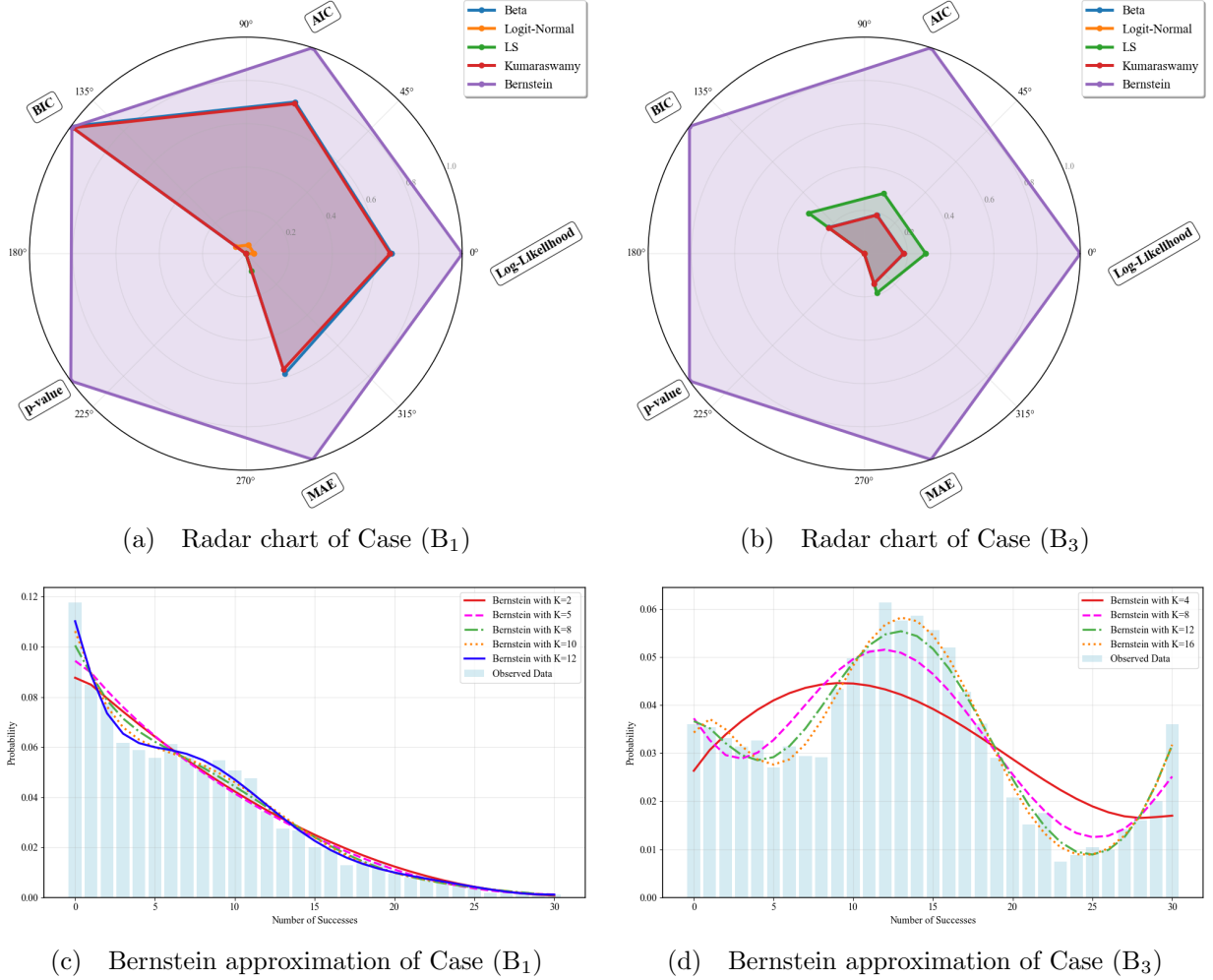


Figure 3: Model comparison for five distributions and Bernstein approximation plots.

4.3 Bayesian inference with DA algorithm

This section will systematically evaluate the performance of Bayesian inference based on DA algorithm in Bern–Bino distribution through numerical simulation. We will evaluate the performance of the posterior mean $\bar{\lambda}$ as a point estimator, the coverage and accuracy of Bayesian credible interval (BCI), the convergence and sampling efficiency of DA algorithm and analyze the impact of different Dirichlet prior settings on the posterior inference.

We will construct multiple simulation scenarios to comprehensively evaluate the performance of the algorithm by controlling the sample size n (Factor 1), the real distribution shape λ (Factor 2) and the Dirichlet prior parameter α (Factor 3). For each scenario deter-

mined by Factor 1, 2 and 3, we will implement DA algorithm with the number of replications $R = 1,000$ and collect evaluation indicators for statistical analysis.

Determine the Bernoulli trial number $m = 15$ in binomial distribution, the following two different real parameter vectors will be considered:

Scenario 1 (S_1): Symmetrical case with $\boldsymbol{\lambda}_{\text{true}} = (0.3, 0.4, 0.3)^\top$;

Scenario 2 (S_2): Skewness with $\boldsymbol{\lambda}_{\text{true}} = (0.25, 0.45, 0.3)^\top$.

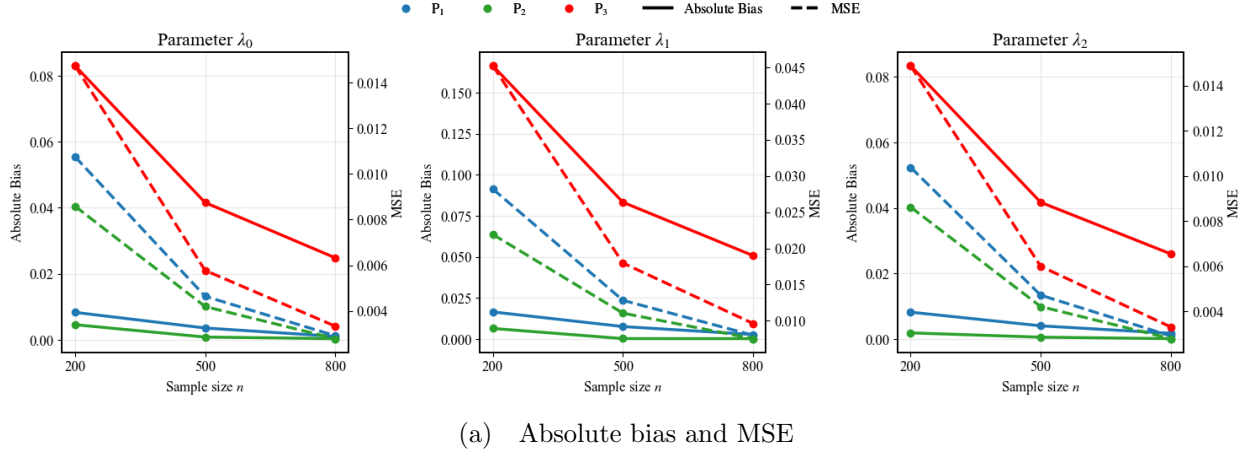
For each scenario, we generate datasets with various sample sizes of $n = 200, 500, 800$ and perform Bayesian inference by using three different Dirichlet priors:

Prior 1 (P_1): Non-informative prior with $\boldsymbol{\alpha} = \mathbf{1}$;

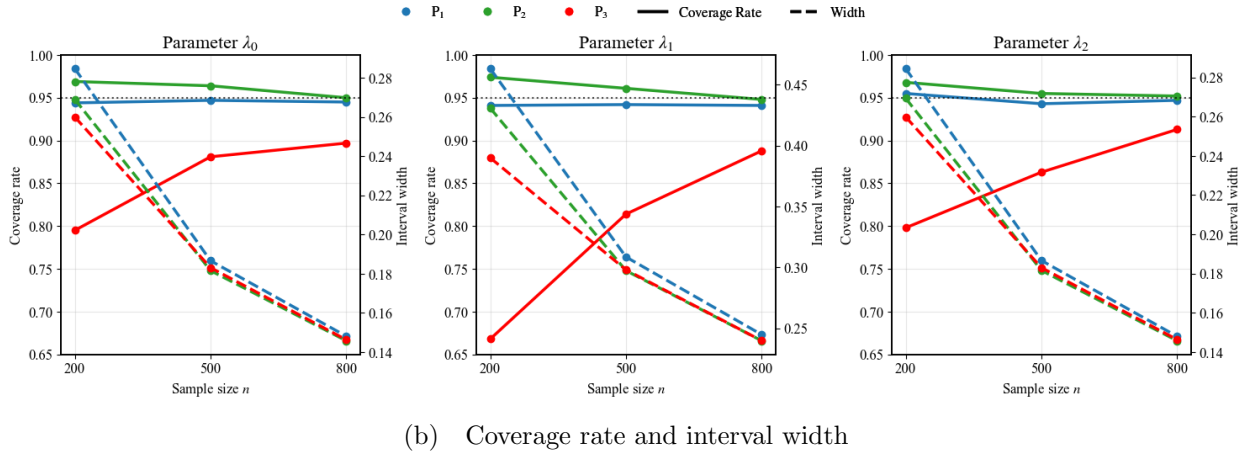
Prior 2 (P_2): Correct prior with $\boldsymbol{\alpha} = 5 \times \boldsymbol{\lambda}_{\text{true}}$;

Prior 3 (P_3): Error prior with $\boldsymbol{\alpha} = (5, 1, 5)^\top$.

For each experimental setup determined by $\{S, n, P, r\}$ in $r = 1, \dots, R$, we run $M = 4$ independent Markov chains. Starting from different random initial values, each chain performs $G = 10,000$ iterations, discards the previous $B = 2,500$ as the burn-in period, and retains the last 7,500 samples for analysis. After the iteration, we combine the four chains to get a posterior sample with a size of 30,000. Based on the combined posterior samples, the posterior mean $\bar{\boldsymbol{\lambda}}$, bias, 95% Bayesian credible interval (BCI), interval width, effective sample size (ESS) and Gelman–Rubin convergent diagnostic statistics (R-hat) can be calculated. By independently repeating R experiments, we can calculate the average of each index, the MSE of the posterior mean and the coverage rate CR of BCI.



(a) Absolute bias and MSE



(b) Coverage rate and interval width

Figure 4: Performance indicators versus sample size under different priors for S_1 .

Figure 4 shows us the absolute bias and MSE of the posterior mean $\bar{\lambda}$, the coverage rate and interval width of 95% BCIs with the sample size n under different priors in Scenario 1. From the figure, we can find that the absolute bias and MSE decrease with the increase of sample size, and the correct prior information can improve the estimation accuracy especially in the case of small samples. For BCI, the interval width will decrease with the increase of the sample size, and the coverage rate of the correct prior (P_2) is the closest to the ideal value, while the coverage rate of the wrong prior (P_3) is obviously low when the n is small, and the impact will weaken with the increase of n . Figure 5 presents the average coverage rate of λ for each prior under various sample sizes by employing the radar chart, where we can intuitively observe the above conclusions.

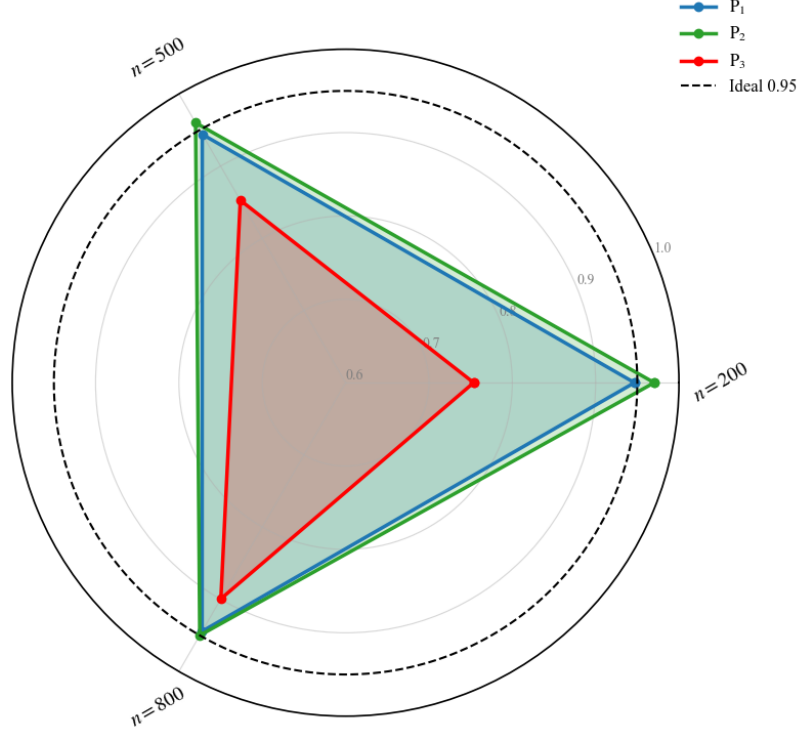


Figure 5: Radar chart of Coverage rate performance for S_1 .

ESS is an indicator to measure the sampling efficiency of MCMC, which converts the MCMC sample chain with autocorrelation into an equivalent independent sample size, which is calculated based on the sum of the sample autocorrelation function (ACF):

$$\text{ESS} = \frac{N}{1 + 2 \sum_{t=1}^{\infty} \rho_t},$$

where N is the total number of MCMC samples and ρ_t is the autocorrelation coefficient of samples with an interval of t . In practice, the sum is usually truncated when ρ_t becomes insignificant. And the Gelman–Rubin statistic $\hat{R}$ (Gelman & Rubin, 1992) is used to diagnose the convergence of MCMC sampling. The statistic is constructed by comparing the inter chain variance (B) and intra chain variance (W) of multiple parallel Markov chains. We calculate the corrected total variance estimate

$$\widehat{\text{Var}}(\boldsymbol{\lambda}) = \frac{N-1}{N}W + \frac{1}{N}B,$$

then the Gelman–Rubin statistic is $\hat{R} = \sqrt{\widehat{\text{Var}}(\boldsymbol{\lambda})/W}$. When the $\hat{R}$ drops below 1.05 even

1.01, it is generally considered that convergence is sufficient.

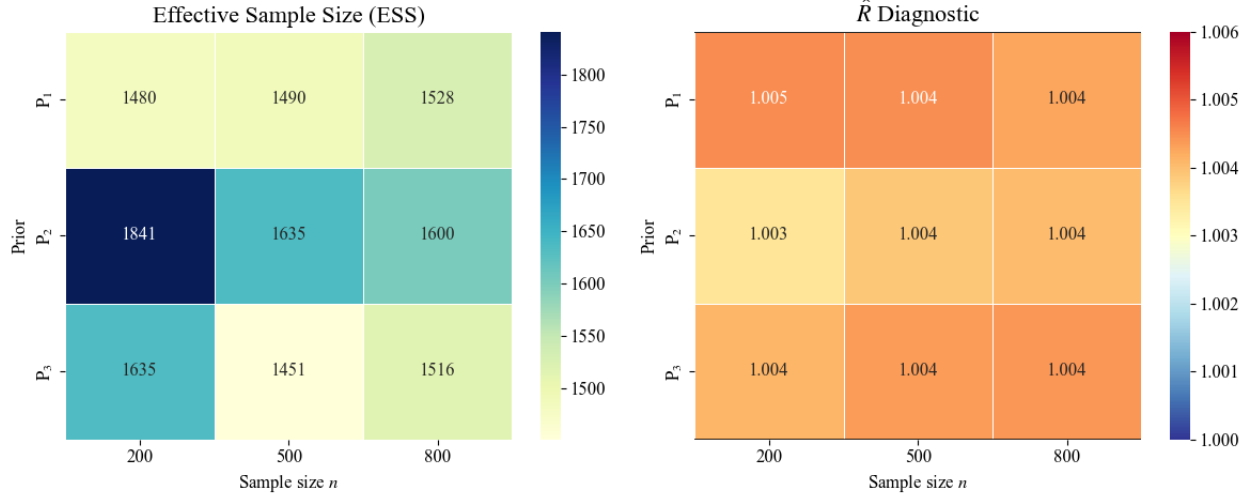


Figure 6: Heat map of DA algorithm efficiency for S_1 .

Figure 6 exhibits the ESS and $\hat{R}$ values of DA algorithm under different sample sizes and priors by adopting heatmap. We can find that the ESS value is greater than 1,000 and $\hat{R}$ is less than 1.01 for each case, which means that the efficiency and reliable convergence of posterior sampling and indicates DA algorithm will not be affected by initial value of sampling. In addition, ESS is significantly larger under the correct the prior and has higher sampling efficiency than other priors. For the scenario 2, we summarize the specific results of Bayesian inference in Table (3), which demonstrates similar conclusions with scenario 1.

Table 3. Influence of different priors on Bayesian inference for S_2 based on $R = 1,000$ replications under various sample sizes

Sample size	Parameter	Prior	Bias	MSE	CR	I-Width	ESS	$\hat{R}$
200	λ_0	P ₁	0.0059	0.0099	0.954	0.277	1685	1.00
		P ₂	−0.0014	0.0084	0.958	0.261	2089	1.00
		P ₃	0.0945	0.0166	0.739	0.258	1887	1.00
	λ_1	P ₁	−0.0136	0.0279	0.959	0.465	1172	1.01
		P ₂	0.0050	0.0229	0.957	0.428	1431	1.00
		P ₃	−0.1890	0.0538	0.589	0.404	1144	1.01
	λ_2	P ₁	0.0076	0.0108	0.955	0.290	1738	1.00
		P ₂	−0.0036	0.0094	0.953	0.271	2172	1.00
		P ₃	0.0946	0.0174	0.761	0.269	1843	1.00
500	λ_0	P ₁	0.0034	0.0042	0.958	0.180	1733	1.00
		P ₂	−0.0010	0.0038	0.964	0.176	1885	1.00
		P ₃	0.0423	0.0057	0.865	0.177	1788	1.00
	λ_1	P ₁	−0.0063	0.0124	0.954	0.305	1208	1.00
		P ₂	0.0025	0.0105	0.967	0.295	1306	1.00
		P ₃	−0.0851	0.0177	0.812	0.295	1200	1.01
	λ_2	P ₁	0.0029	0.0048	0.950	0.188	1783	1.00
		P ₂	−0.0015	0.0043	0.956	0.183	1946	1.00
		P ₃	0.0427	0.0061	0.867	0.185	1796	1.00
800	λ_0	P ₁	0.0017	0.0027	0.944	0.143	1767	1.00
		P ₂	−0.0016	0.0026	0.952	0.141	1856	1.00
		P ₃	0.0303	0.0034	0.872	0.142	1807	1.00
	λ_1	P ₁	−0.0028	0.0077	0.947	0.242	1229	1.00
		P ₂	0.0025	0.0073	0.939	0.237	1287	1.00
		P ₃	−0.0586	0.0103	0.857	0.237	1223	1.00
	λ_2	P ₁	0.0011	0.0029	0.946	0.149	1813	1.00
		P ₂	−0.0009	0.0028	0.941	0.147	1912	1.00
		P ₃	0.0283	0.0035	0.900	0.147	1827	1.00

5. Real data analysis

6. Conclusion and Discussion

Supplementary Materials

Acknowledgement

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Reference

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A. Some Technical Derivations

A.1 Asymptotic normality theorem for MLE

Suppose that $\{X\}_{i=1}^n \stackrel{\text{iid}}{\sim} F(x|\theta)$, the likelihood function be expressed as $\mathcal{L}_n(\theta)$. Then we calculate the MLE $\hat{\theta}_n$. To study the consistency and asymptotic normality of $\hat{\theta}_n$, we need a laundry list of assumptions, which has come to be known as the Cramér–Rao conditions (Lehmann & Casella 1998 and Lehmann 1999).

Theorem 2 (Cramér–Rao conditions)

Let $X_1, \dots, X_n$ be i.i.d. r.v. with density $f(x|\theta)$ (or discrete case with probability $\Pr(x|\theta)$) and *cumulative distribution function* (cdf) $F(x|\theta)$. In order to study the consistency and asymptotic normality of MLE $\hat{\theta}_n$ of parameter θ , we assume the following conditions:

(A1) Identifiability, that is, $F(x|\theta_1) = F(x|\theta_2) \iff \theta_1 = \theta_2$;

(A2) $\theta \in \Theta$ is an open interval in $\mathbb{R}^d$;

(A3) The observations are $X = (X_1, \dots, X_n)$, where the X_i are i.i.d. either with probability density $f(x|\boldsymbol{\theta})$ which will be assumed to be continuous in x (the continuous case), or discrete with probabilities $\Pr(x|\boldsymbol{\theta})$ (the discrete case).

(A4) The support $\mathbb{S} = \{x : f(x|\boldsymbol{\theta}) > 0\}$ is free from $\boldsymbol{\theta}$;

(A5) $\forall x \in \mathbb{S}$, $f(x|\boldsymbol{\theta})$ or $\Pr(x|\boldsymbol{\theta})$ is differentiable with respect to $\boldsymbol{\theta}$, that is, the likelihood function is smooth as a function of the parameter.

Let $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta}$ be the true value of $\boldsymbol{\theta}$. Then there exists a sequence $\hat{\boldsymbol{\theta}}_n$ such that $\hat{\boldsymbol{\theta}}_n \rightarrow \boldsymbol{\theta}_0$ in probability. Under the condition of consistency, Assumption (A6) and (A7) provide sufficient conditions for the asymptotic normality of $\hat{\boldsymbol{\theta}}_n$.

(A6) For all $x \in \mathbb{S}$ (specified in (A4)), the density $f(x|\boldsymbol{\theta})$ is three times differentiable with respect to $\boldsymbol{\theta}$ and the third derivative is continuous. The corresponding integral (or sum in discrete case) of partial derivative

$$\int \frac{\partial^3 f(x|\boldsymbol{\theta})}{\partial \theta_r \partial \theta_s \partial \theta_t} dx$$

can be obtained by differentiating under the integral sign (or sum);

(A7) There exists a positive number $c(\boldsymbol{\theta}_0)$ and a function $M_{\boldsymbol{\theta}_0}(x)$ such that

$$\left| \frac{\partial^3}{\partial \theta_r \partial \theta_s \partial \theta_t} \log f(x|\boldsymbol{\theta}) \right| \leq M_{\boldsymbol{\theta}_0}(x) \quad \text{for all } x \in \mathbb{S}, \quad \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| < c(\boldsymbol{\theta}_0),$$

and $E_{\boldsymbol{\theta}_0}(M_{\boldsymbol{\theta}_0}(X)) < \infty$.

Then any consistent sequence $\hat{\boldsymbol{\theta}}_n$ satisfies

$$\sqrt{n} \left(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0 \right) \xrightarrow{D} N(0, \mathbf{J}^{-1}(\boldsymbol{\theta}_0)),$$

where $\mathbf{J}(\boldsymbol{\theta}) = [J_{ij}(\boldsymbol{\theta})]$ with

$$J_{ij}(\boldsymbol{\theta}) = -E \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log \mathcal{L}(\boldsymbol{\theta}|\mathbf{X}) \right]$$

is the Fisher information matrix, here $\mathbf{X} = (X_1, \dots, X_n)$. ||

A.2 Validation of Cramér–Rao conditions for Bern–Bino distribution

Suppose that $\{X_i\}_{i=1}^n \stackrel{\text{iid}}{\sim} \text{Bern-Bino}(m, K, \boldsymbol{\lambda})$ with $\boldsymbol{\lambda} \in \mathbb{T}_{K+1}^{\text{open}}$, which satisfies Assumption (A3) in Theorem (2) with discrete case. And the pmf of X_i is given by

$$\text{Bern-Bino}(x_i|m, K, \boldsymbol{\lambda}) \stackrel{(2.2)}{=} \binom{m}{x_i} \sum_{k=0}^K a_{ik} \lambda_k, \quad x_i \in \mathbb{N}_{0:m},$$

where the parameter $\boldsymbol{\lambda}$ is identifiability (reason in Section 1.2) and $\mathbb{T}_{K+1}^{\text{open}}$ is a open set in $\mathbb{R}^{K+1}$, which satisfies Assumption (A1), (A2). The support $\mathbb{S}_{X_i} = \mathbb{N}_{0:m}$ is free from $\boldsymbol{\lambda}$, therefore, the Assumption (A4) holds. The log-likelihood function of $\boldsymbol{\lambda}$ is

$$\ell(\boldsymbol{\lambda}|Y_{\text{obs}}) \stackrel{(2.6)}{=} \sum_{i=1}^n \log \binom{m}{x_i} + \sum_{i=1}^n \log \left(\sum_{k=0}^K a_{ik} \lambda_k \right) \quad \text{with } \boldsymbol{\lambda} \in \mathbb{T}_{K+1}^{\text{open}},$$

which is a smooth function for $\boldsymbol{\lambda}_{-0}$ with partial derivatives

$$\begin{aligned} \frac{\partial \ell(\boldsymbol{\lambda}|Y_{\text{obs}})}{\partial \lambda_r} &= \sum_{i=1}^n \frac{a_{ir} - a_{i0}}{\sum_{l=0}^K a_{il} \lambda_l} \quad \text{and} \\ \frac{\partial^3 \ell(\boldsymbol{\lambda}|Y_{\text{obs}})}{\partial \lambda_r \partial \lambda_s \partial \lambda_t} &= \sum_{i=1}^n \frac{2(a_{ir} - a_{i0})(a_{is} - a_{i0})(a_{it} - a_{i0})}{(\sum_{l=0}^K a_{il} \lambda_l)^3}, \quad r, s, t = 1, \dots, K. \end{aligned}$$

Because pmf is a linear function of λ_k , which has good properties, so the log-likelihood function is third times differentiable and the derivative function is continuous. In the neighbor of true value $\boldsymbol{\lambda}_{-0, \text{true}}$, we can choose a small enough constant $c(\boldsymbol{\lambda}_{-0, \text{true}})$ such that all $\lambda_k \geq \delta > 0$. Therefore, we obtain the upper bound

$$\left| \frac{\partial^3 \ell(\boldsymbol{\lambda}|Y_{\text{obs}})}{\partial \lambda_r \partial \lambda_s \partial \lambda_t} \right| \leq \frac{2c_{\max}^3}{(\delta c_{\min})^3}$$

is a constant that does not depend on x or the indicators r, s, t , where $c_{\max} = \max\{|a_{ik} - a_{i0}| \mid i = 1, \dots, n, k = 0, 1, \dots, K\}$ and $c_{\min} = \min_{1 \leq i \leq n, 0 \leq k \leq K} a_{ik}$ are two constants. Then the three times partial derivative has upper bound function $M_{\boldsymbol{\lambda}_{-0, \text{true}}}(x) = 2c_{\max}^3/(\delta c_{\min})^3$ with finite moment $E[M_{\boldsymbol{\lambda}_{-0, \text{true}}}(X)] = 2c_{\max}^3/(\delta c_{\min})^3 < \infty$. Therefore, we verified Assumption (A5), (A6) and (A7) hold. In conclusion, for $\boldsymbol{\lambda}_{-0} \in \mathbb{T}_K^{\text{open}}$, Bernstein–Binomial distribution satisfy the Cramér–Rao conditions.

B. Some other results in article

Table 4. Influence of different priors on Bayesian inference for S_1 based on $R = 1,000$ replications under various sample sizes

Sample size	Parameter	Prior	Bias	MSE	CR	I-Width	ESS	$\hat{R}$
200	λ_0	P ₁	0.0083	0.0108	0.944	0.285	1657	1.00
		P ₂	−0.0046	0.0086	0.969	0.269	2076	1.00
		P ₃	0.0830	0.0147	0.795	0.260	1897	1.00
	λ_1	P ₁	−0.0165	0.0282	0.941	0.463	1116	1.01
		P ₂	0.0065	0.0219	0.974	0.431	1368	1.00
		P ₃	−0.1660	0.0452	0.668	0.390	1111	1.01
	λ_2	P ₁	0.0082	0.0104	0.955	0.284	1667	1.00
		P ₂	−0.0019	0.0086	0.968	0.269	2079	1.00
		P ₃	0.0834	0.0149	0.798	0.260	1897	1.00
500	λ_0	P ₁	0.0036	0.0046	0.947	0.187	1668	1.00
		P ₂	0.0009	0.0042	0.964	0.182	1839	1.00
		P ₃	0.0416	0.0058	0.881	0.183	1632	1.00
	λ_1	P ₁	−0.0076	0.0128	0.942	0.309	1134	1.01
		P ₂	−0.0003	0.0111	0.961	0.297	1234	1.00
		P ₃	−0.0833	0.0180	0.814	0.298	1082	1.01
	λ_2	P ₁	0.0040	0.0047	0.943	0.187	1668	1.00
		P ₂	−0.0006	0.0042	0.955	0.182	1833	1.00
		P ₃	0.0417	0.0060	0.863	0.183	1639	1.00
800	λ_0	P ₁	0.0011	0.0029	0.945	0.148	1709	1.00
		P ₂	0.0004	0.0028	0.950	0.146	1795	1.00
		P ₃	0.0249	0.0033	0.897	0.146	1702	1.00
	λ_1	P ₁	−0.0028	0.0080	0.941	0.245	1161	1.01
		P ₂	−0.0003	0.0075	0.948	0.240	1212	1.01
		P ₃	−0.0508	0.0096	0.888	0.240	1139	1.01
	λ_2	P ₁	0.0017	0.0029	0.947	0.148	1715	1.00
		P ₂	−0.0001	0.0028	0.952	0.146	1794	1.00
		P ₃	0.0260	0.0033	0.913	0.146	1707	1.00